

SiRho.

F R A M E S

*Adaptive Skating,
Engineered for Performance.*

Sinai Rhodes



Voronoi Structured Mesh

The frame incorporates a Voronoi lattice to significantly reduce weight without compromising strength. This optimised structure enhances agility while maintaining durability, ensuring it can withstand high-impact movements such as jumps.

Suspension Block

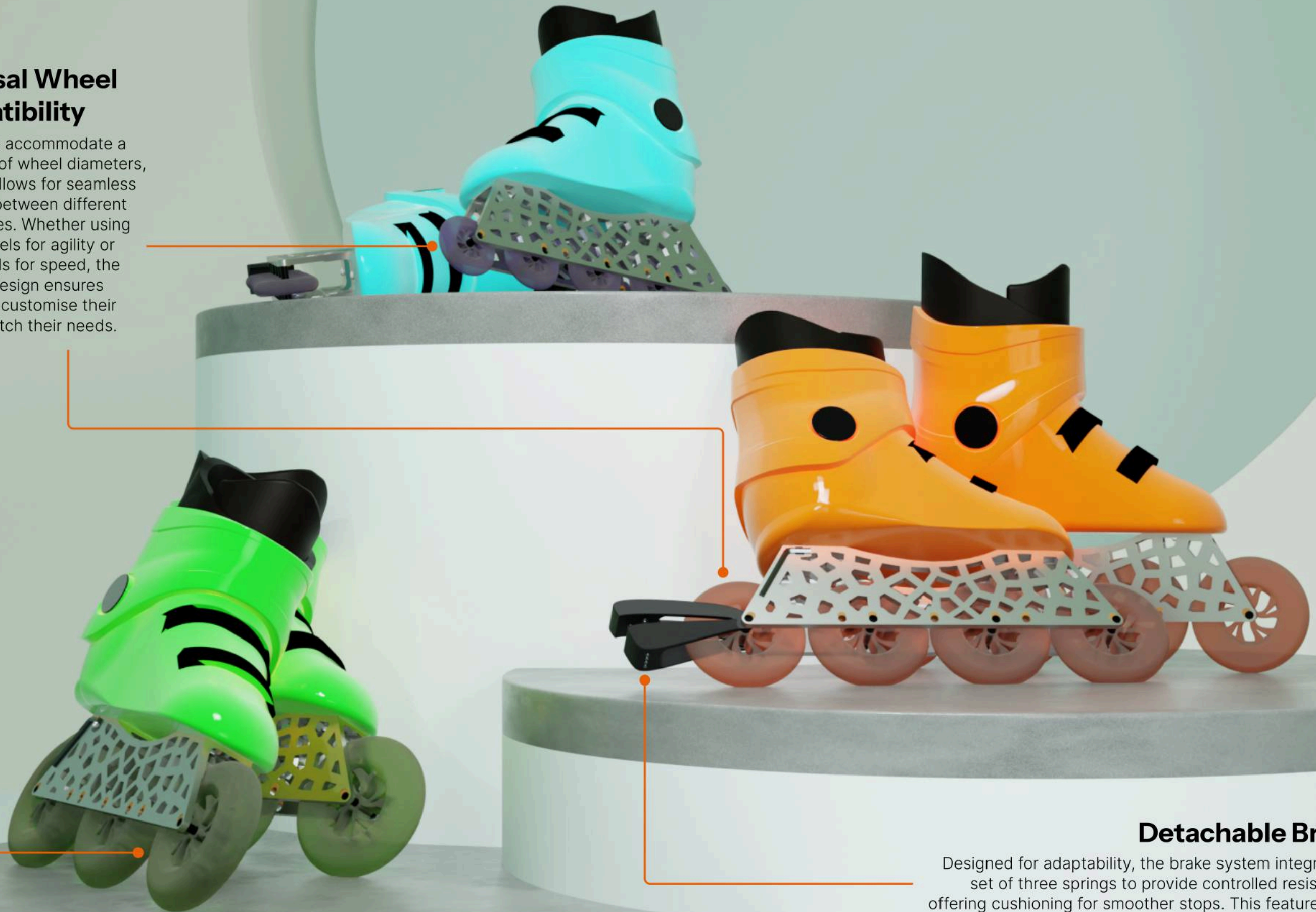
Featuring a gyroid mesh structure, the 3D-printed TPU suspension block effectively absorbs shocks and vibrations. This design enhances ride comfort and stability across various surfaces while maintaining flexibility and durability under repeated compression.

Allen Key

Securely held in place with neodymium magnets, the Allen key is integrated at a 30-degree angle within the frame, ensuring easy access while maintaining a sleek design. This eliminates the need to carry external tools, enhancing convenience for skaters.

Universal Wheel Compatibility

Designed to accommodate a wide range of wheel diameters, this frame allows for seamless transitions between different skating styles. Whether using smaller wheels for agility or larger wheels for speed, the adaptable design ensures skaters can customise their setup to match their needs.



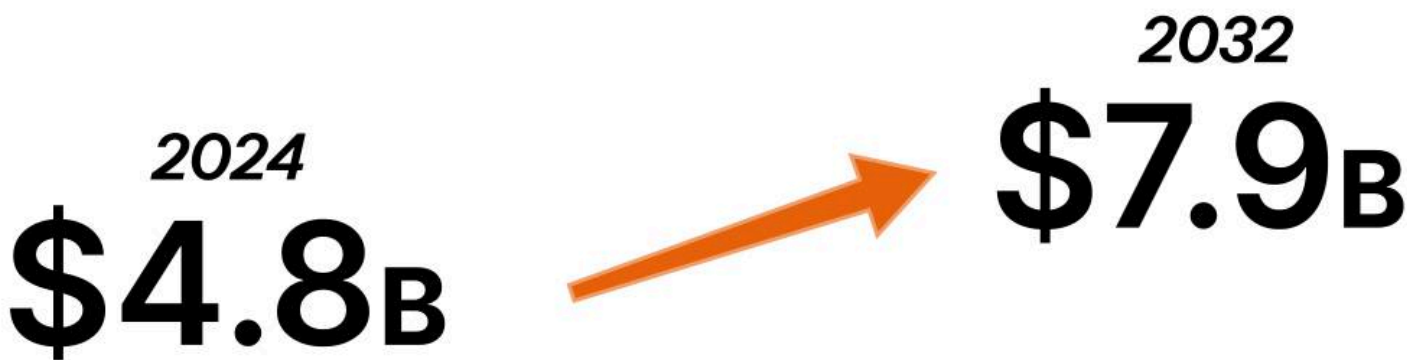
Detachable Brake

Designed for adaptability, the brake system integrates a set of three springs to provide controlled resistance, offering cushioning for smoother stops. This feature helps new skaters gradually develop braking confidence while reducing over-reliance on the brake for stopping.

Context.

Market Size.

The global roller skate market is experiencing significant growth, driven by rising interest in recreational activities and fitness pursuits. Over the next 7 years the market is projected to increase by approximately 65% with a Compound Annual Growth Rate (CAGR) of 6.2%. [1]



Trends.

The roller skate market is rapidly evolving, driven by growing recreational interest, fitness trends, and continuous product innovation. Social media platforms like Instagram and TikTok have contributed to this growth by showcasing creative skate styles, driving popularity of the sport.

Recreation

The recent revival across diverse age groups is driven by the enjoyable, social lifestyle. The Roller Skating Association reported an 11.2% increase in roller sports participation in 2022.

[2]

Sustainability

Consumers are increasingly prioritising sustainability. As a result, the market is shifting towards sustainable manufacturing methods of skate products to reduce environmental impact.

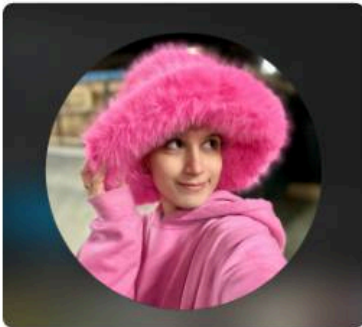
[3]

Customisation

Consumers increasingly value personalised skate setups, with modular components, adjustable features, and aesthetic flexibility emerging as key market preferences.

[3]

Interview.



Ilaria Di Maio

Manager @ Slick Willies
Skating for 15 years

What’s the most common complaint or issue you hear from customers about skates?

“There are too many options... people come in and want to get some skates and are confused what to get”

Do customers ask about skates that can adapt to multiple styles?

“Definitely, I’ve got so many different skates... especially for people who want to do aggressive, freestyle and slalom it’s a lot”

Are customers typically experienced or beginners?

“It’s a mix... the more experienced skaters normally come in knowing what they want but beginners need a lot of advice.”

Pain Points.

From Ilaria's interview and secondary research, several key pain points have emerged in the roller skate market, reflecting challenges skaters face in terms of performance and general use.

Too Much Choice

Lacks Customisation

Heavy

Unnatural Brake Placement

Expensive

Lacks Versatility

Comfortability

Maintenance

Shock Absorption

Design Requirements.

The design requirements address key market trends and user pain points, focusing on customisation, modularity, and comfort. By improving weight, versatility, and durability, the goal is to enhance the skating experience for all users with a practical and high-performance solution.

Lightweight & Durable

Utilising strong yet lightweight materials to enhance performance without compromising durability or longevity.

Modular Adaptability

A frame that accommodates removable components, such as brakes, to transition between different skating styles seamlessly.

Shock Absorption

Integrated impact-reducing features to minimise vibrations and improve comfort across various skating surfaces.

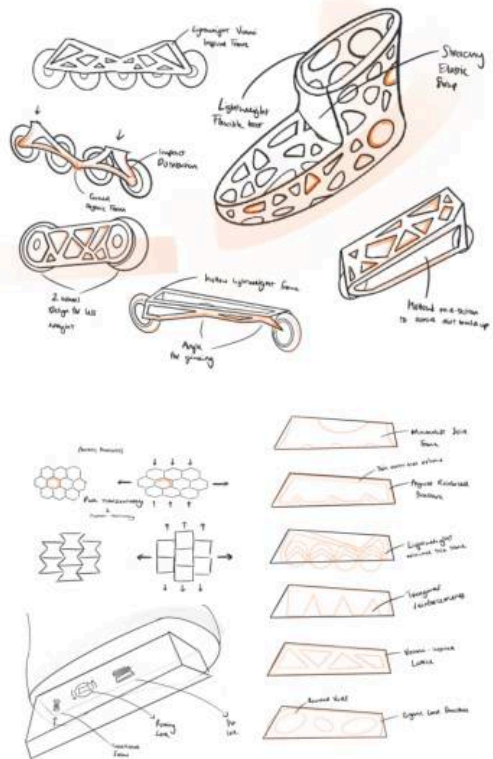
Initial Ideation.

Moodboard.

The moodboard explores biomimetic and parametric design influences, showcasing lightweight lattice structures, advanced material textures, and high-performance geometries.

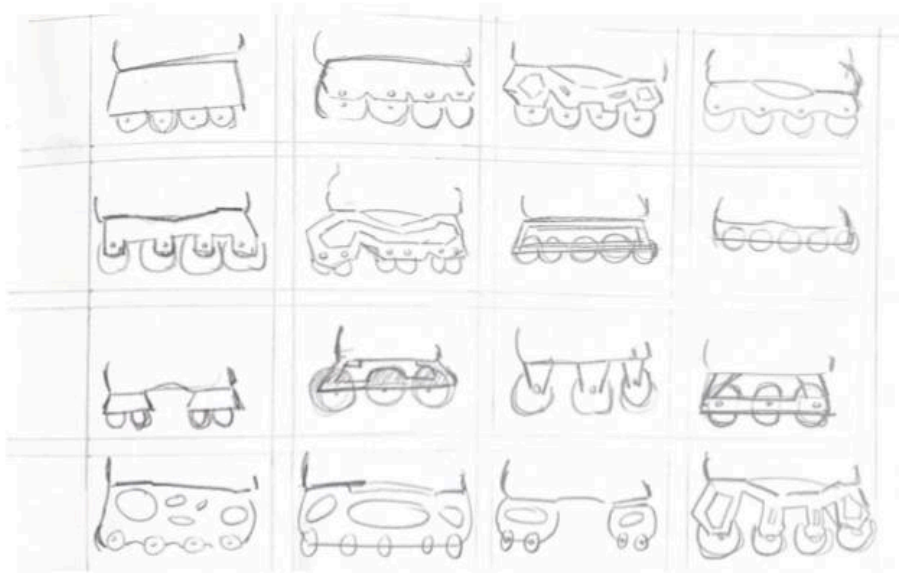


Initial Sketching.



Form Exploration.

I explored a variety of frame forms, spending 30 seconds on each sketch to experiment with different profiles. This helped evaluate aesthetic appeal and functionality, to identify the best design for performance and versatility.



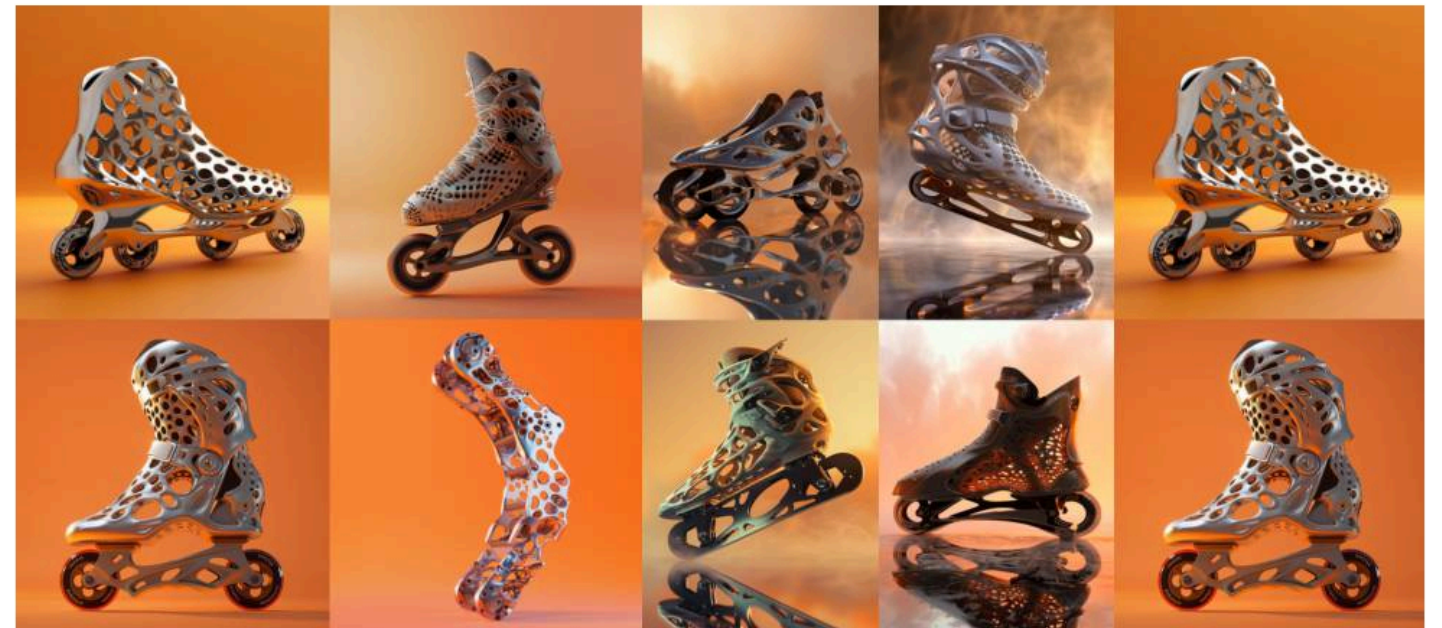
Morphological Analysis.

Morphological analysis was used to explore different functional and aesthetic aspects of the skate frame. I broke down the design into key categories, including side profile, cutout patterns, mounting system, brake form, and wheel attachment, to systematically generate diverse concept variations. After evaluating the options, I selected the most promising configurations to take forward into the prototyping phase.

FUNCTION	CONCEPTS				
	1	2	3	4	5
SIDE PROFILE	WIZARD 	CURVED 	ANGLED 	ORGANIC 	SCULPTED
CUTOUT PATTERNS	ORGANIC 	VORONOI 	HEXAGONAL 	SLITS 	GEOMETRIC
MOUNTING SYSTEM	SCREEN 	SNAP FIT 	MAGNET 	SLIDING 	THrust
BRAKE FORM					
WHEEL ATTACHMENT	AXLE 	SPRING SWIP. 	VARIABLE 	SLOTS 	MAGNET

Midjourney Exploration.

Using midjourney I explored a range of skate concepts, focusing on organic, high-performance structures inspired by lightweight architecture. Each image was generated with minimal prompts and analysed for its form, function, and visual language. This iterative sketching process helped inspire my final frame design, balancing aesthetics with structural clarity.



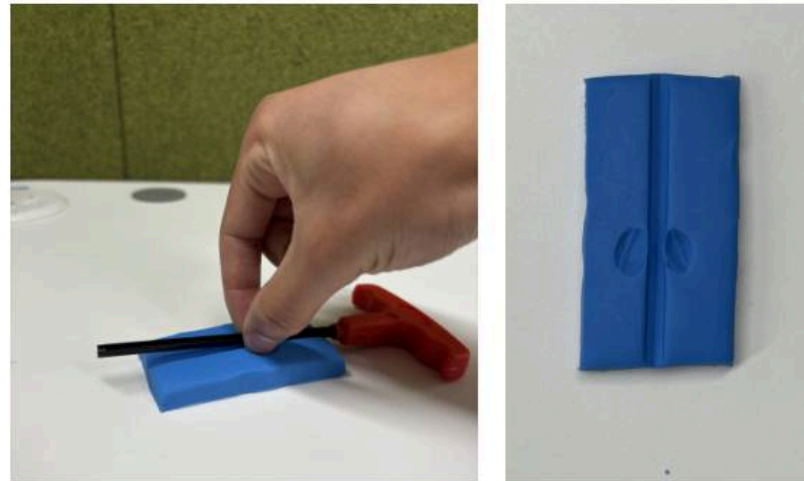
Prototyping.

Allen Key Storage.

Current frames require a set a of Allen keys of 2 or more to fully tighten and/or change components of the skate including bearings, wheels and frames. These are not only easy to use but are dangerous to carry on your person when skating and so I wanted to explore how the tool could be integrated into the existing rollerblade infrastructure to eliminate this risk.

To explore this I began by analysing the the impression made of a common bulky Allen key in modelling clay, in addition analysing the impression of picking up the frame.

I tested various Allen key storage designs and chose a lofted off-centre hexagonal slot for secure retention and easy access. This design mimics a finger press while maintaining the hexagonal shape, ensuring stability under vibration.



Skating Style Accomodation.

During prototyping, various wheel diameters were tested to ensure adaptability across different skating styles.

- **55mm-60mm** wheels, commonly used in aggressive skating, offer maximum control for grinds and tricks but limit speed.
- **72mm-80mm** wheels, typical in urban and freestyle skating, provide a balance between agility and stability.
- **90mm-110mm** wheels, often seen in freeride and marathon skating, enhance speed and glide efficiency but require a longer frame for stability.

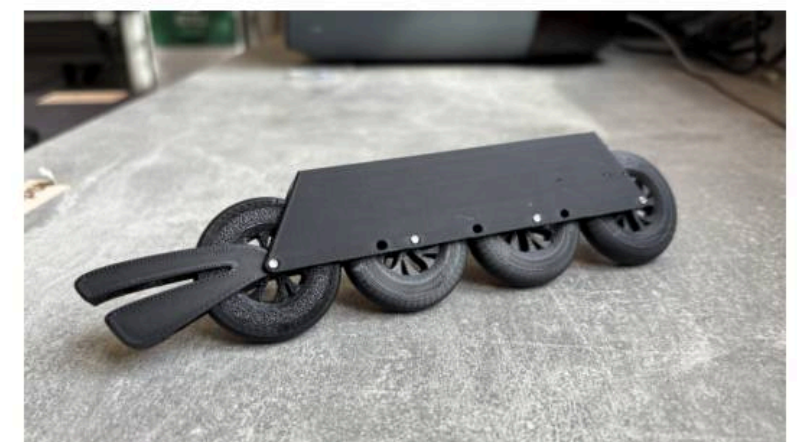
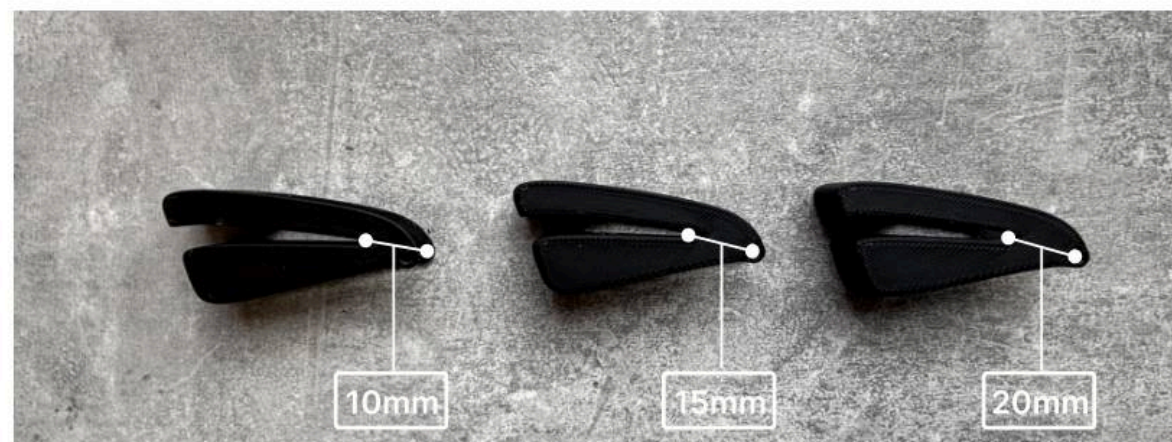
By designing the frame to accommodate this full range, the prototype ensures versatility for multiple skating disciplines.



Brake Exploration.

To improve beginner confidence and safety, I prototyped a spring-integrated brake system designed to gradually reduce reliance over time. I experimented with varying hinge thickness deflections (10mm, 15mm, 20mm with a scale factor of 5:1) to test resistance and cushioning selecting 15mm as the most optimal. The aim was to minimise the sharp jolts caused by traditional fixed brakes.

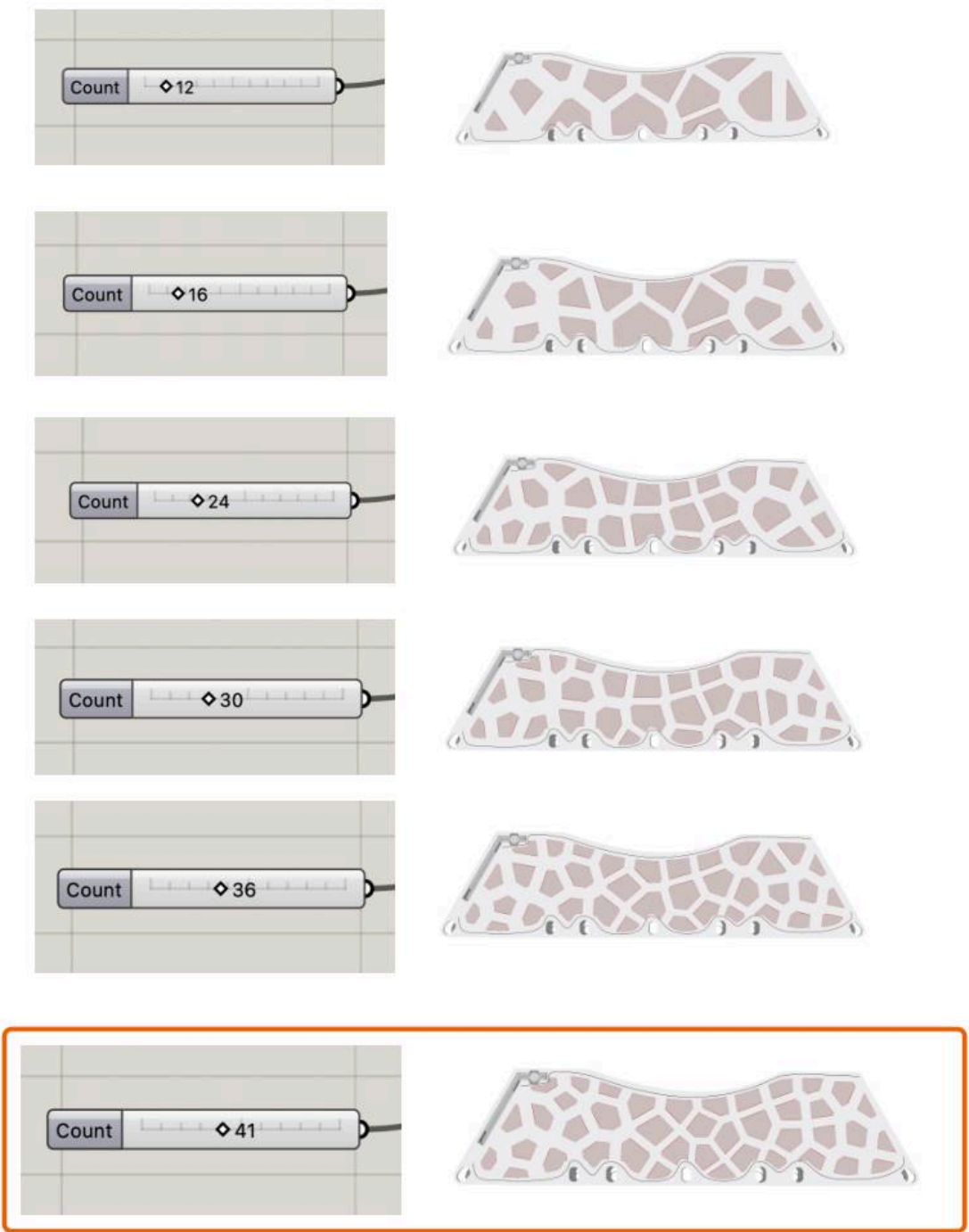
The initial concept relied on a compliant flexing mechanism, but to better absorb extreme forces and repeated use, a set of embedded springs was added to the final design.



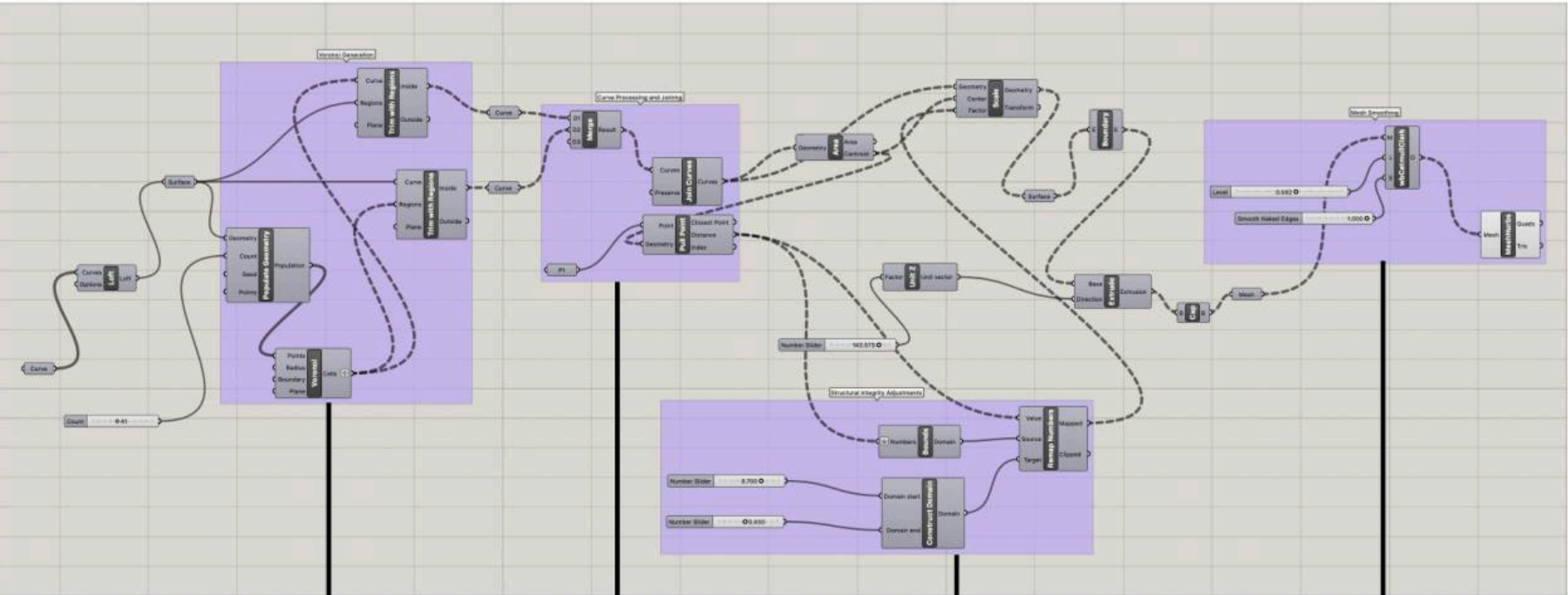
CAD Development.

Voronoi Development.

The Voronoi structure was adjusted iteratively to optimise weight reduction while maintaining structural integrity. The cell count was varied until no Voronoi cells overlapped with axle holes, as this would compromise strength and increase the risk of breakage during high-impact activities like jumps. Lower cell counts resulted in larger, rigid voids, while higher counts increased complexity and potential flexibility. This refinement ensured a balance between lightweight design and durability, enhancing performance without structural compromise.



Grasshopper.



Voronoi Generation

The Voronoi pattern was applied to introduce lightweight voids within the frame while maintaining its strength. The number of Voronoi cells was varied to explore different densities, ensuring no cells were positioned above axle holes to prevent structural weaknesses. This phase was critical in achieving a balance between rigidity and flexibility.

Curve Processing

Once the Voronoi cells were generated, they were cleaned, trimmed, and joined to ensure smooth integration with the frame's boundary. The process involved refining the pattern to fit within the frame's geometry while maintaining a structurally sound and visually cohesive design.

Structural Integrity

To ensure the frame could withstand high-impact forces such as jumps, the Voronoi structure was assessed for load distribution and stress points. The density of cells was adjusted to maintain a strong, rigid structure where necessary while removing excess material in less critical areas. This process optimised durability without unnecessary weight.

Mesh Smoothing

To achieve a manufacturable and visually refined design, the frame was converted into a smooth, high-resolution mesh. This step ensured that the Voronoi edges were rounded and uniform, reducing stress concentrations and improving the overall aesthetic and structural performance of the frame.

CMF.

Colours.

The chosen colour palette reflects a bold, high-energy aesthetic, appealing to skaters who value both performance and personal expression. The vibrant hues—fluorescent green, deep orange, and electric purple—enhance visibility for safety while reinforcing a modern, futuristic look. The contrast between the metallic frame and bright boot colours highlights the engineering precision of the design, ensuring it stands out in both urban and competitive skating environments.



#E8630C



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Materials and Finishes.

Axles and Screws

Precision **CNC-machined Stainless Steel (Grade 304)** ensures strength and corrosion resistance. A polished finish minimises friction and enhances durability.

Frame

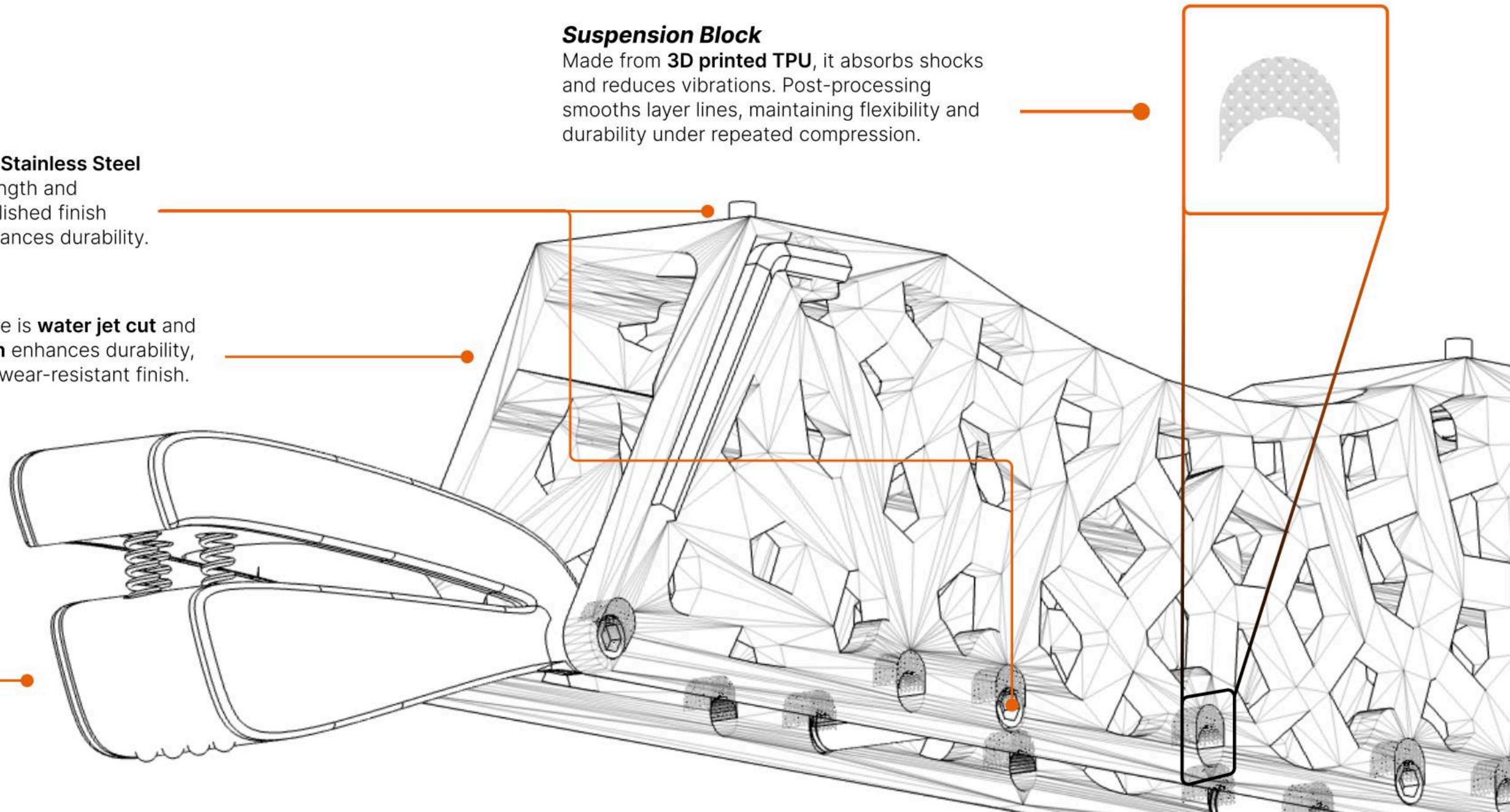
Made from **Aluminium 6061-T6**, the frame is **water jet cut** and **CNC machined** for precision. **Anodisation** enhances durability, prevents oxidation, and provides a sleek, wear-resistant finish.

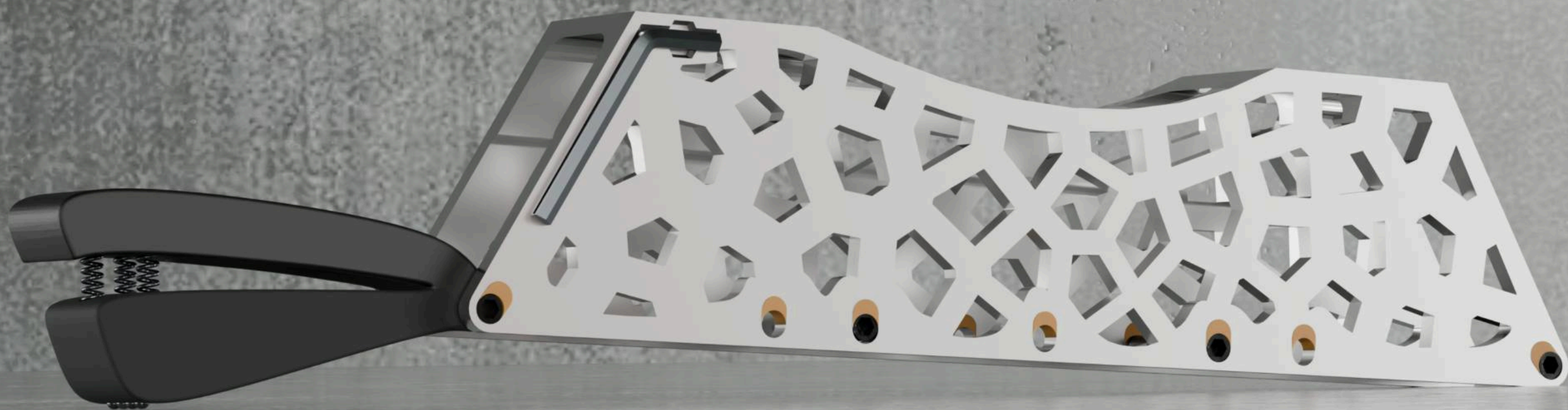
Brake

Crafted from **Polyurethane**, the brake is **injection moulded** for consistency. Its **matte black finish** reduces glare, while the material's durability ensures long-lasting performance.

Suspension Block

Made from **3D printed TPU**, it absorbs shocks and reduces vibrations. Post-processing smooths layer lines, maintaining flexibility and durability under repeated compression.





References & Appendix.

1. Business Research Insights (2024) Roller Skating Market Size, Share, Growth and Industry Analysis 2024-2029. Available at: <https://www.businessresearchinsights.com/market-reports/roller-skating-market-117237> (Accessed: 11 March 2025).
2. Roller Skating Association International (2024) Roller skating industry trends and insights. Available at: <https://www.rollerskating.com/blog/archives/07-2024> (Accessed: 11 March 2025).
3. Verified Market Reports (2024). The Growing Top 7 Trends in the Roller Skate Market. Available at: <https://www.verifiedmarketreports.com/blog/the-growing-top-7-trends-in-the-roller-skate-market/> (Accessed: 11 March 2025).

MidJourney Prompts.

/imagine prompt: A sleek rollerblade frame, floating in a minimalist studio, entirely covered in organic voronoi mesh design, made from brushed silver metal. Isolated on an orange-tinted gradient background. High-detail close-up showing smooth curves and frame structure. Created Using: CAD render, Fusion 360, Voronoi algorithm modeling, soft rim lighting, studio HDRI, stainless steel texture mapping, photoreal quality, technical focus --ar 16:9 --v 6.0

imagine prompt: Rollerblade frame suspended mid-air, intricate organic voronoi mesh pattern wrapping around the metallic surface. No boot, just the frame, glowing softly against a blurred orange light backdrop. Perspective from above with dramatic shadow play. Created Using: 3D printing mockup, solidworks modeling, ambient occlusion render, brushed aluminum textures, soft directional lighting, industrial product style, sharp depth of field --ar 3:2 --v 6.0

/imagine prompt: A futuristic rollerblade frame design, no wheels or boot, just the structure, detailed with flowing organic voronoi mesh. Hovering above a reflective surface, orange-tinted foggy background. Photorealistic and professional. Created Using: Blender cycles, parametric modeling, 3D CAD render pass, soft HDR light dome, titanium finish shader, focus on structural flow --ar 2:3 --v 6.0